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# On the numerical treatment of moving bottlenecks

Carlos F. Daganzo *, Jorge A. Laval

*Institute of Transportation Studies and Department of Civil and Environmental Engineering, University of California, 109 McLaughlin Hall #1720, Berkeley, CA 94720, USA*

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**Abstract**

This paper shows how moving obstructions in (kinematic wave) traffic streams can be modeled with "off-the shelf" computer programs. It shows that if a moving obstruction is replaced by a sequence of fixed obstructions at nearby locations with the same "capacity", then the error in vehicle number converges uniformly to zero as the maximum separation between the moving and fixed bottlenecks is reduced. This result implies that average flows, densities, accumulations and delays can be predicted as accurately as desired with this method. Thus, any convergent finite difference scheme can be used to model moving bottlenecks. The approach can be used with non-concave fundamental diagrams and multiple bottlenecks, even if they pass each other. Examples are given. It is assumed that the bottleneck trajectories are exogenous to the model. However, by introducing suitable car-following laws and interaction rules, slow trucks and busses embedded in the traffic stream can be modeled endogenously.
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*Keywords:* Traffic flow; Moving bottlenecks; Numerical methods

## 1. Introduction

Our experience as drivers tells us that a truck on an uphill, a vehicle changing lanes or a vehicle accelerating while entering the freeway can affect significantly freeway traffic streams. Empirical findings confirm this intuition. It is known, for example, that lane changes are an important contributor to the formation of stop-and-go waves (Mauch and Cassidy, 2002), and that slow trucks contribute to the formation of sag bottlenecks (Koshi et al., 1992). Controlled experiments

* Corresponding author. Tel.: +1-510-642-3853; fax: +1-510-643-8919.
*E-mail address:* daganzo@ce.berkeley.edu (C.F. Daganzo).

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show that the effects of moving obstructions are indeed significant (Muñoz and Daganzo, 2002). This suggests that "moving bottlenecks" should be incorporated into practical traffic models.

Gazis and Herman (1992) were first to recognize the moving bottleneck problem and to address some of its issues. Later, moving obstructions were introduced into the kinematic wave (KW) theory of Lighthill and Whitham (1955), and Richards (1956), and this led to a complete theory of moving bottlenecks (Newell, 1998). Because this theory did not fully agree with observation, it was revised to improve its predictions (Muñoz and Daganzo, 2002). With these models it is possible to predict the effects of any set of moving obstructions on a KW traffic stream. Unfortunately, a general way of discretizing these models has not yet been found.

Lebacque et al. (1998) have tackled the special case where bottlenecks travel at a fixed common speed that is an integer submultiple of the free-flow speed by recasting the problem in a frame of reference moving with the bottlenecks. Subsequently, Giorgi et al. (2002) relaxed the single speed assumption with an extension of Godunov's scheme where the passing rate in the neighborhood of the bottleneck is modeled in a fixed frame of reference. However, this model still requires the bottleneck speed to be an integer submultiple of the free-flow speed and does not allow for bottleneck trajectories to cross. Thus, existing methods are not yet flexible enough for application to general problems where the bottlenecks may follow arbitrary trajectories; e.g., where trucks with differing characteristics encounter and cause queue(s) on a sag vertical curve.

Here we propose an alternative approach that can be used with multiple bottlenecks and no restrictions on their trajectories. This is accomplished by replacing the exact problem with an approximate problem whose solution turns out to be close to the desired solution. We will model the time-space $(t,x)$ trajectories of the moving obstructions $\phi(t)$ approximately, by step functions $\phi^{(a)}(t)$ with steps equal to the lattice spacing, $\Delta x$; see Fig. 1. Although it is possible to choose $\phi^{(a)}(t) \to \phi(t)$ by letting $\Delta x \to 0$, a difficulty arises because it is not generally possible to match the flows and densities of the solutions on both sides of $\phi$ and $\phi^{(a)}$. To see this note that if a bottleneck $\phi$ is in an active steady state (i.e., it is holding back a queue) while traveling at a speed $v = \mathrm{d}\phi/\mathrm{d}t > 0$, then the two flow-density $(q,k)$ states directly upstream and downstream of the bottleneck must satisfy the "shock condition" $\Delta q/\Delta k = v$. This means that on the $(q,k)$ plane they must be on a "capacity line" with slope $v$; see points $U$ and $D$ on Fig. 2. [1] Thus, the flow in these states must be different. Yet, for a bottleneck with zero speed, such as any step of $\phi^{(a)}$, the difference in flows $\Delta q^{(a)}$ must be zero, as shown by points $U^{(a)}$ and $D^{(a)}$ of the figure. Obviously, the approximate solution cannot precisely match the true flows on both sides of an active bottleneck. Fortunately, this difficulty is not fatal. We will show that if the capacity of the approximate moving bottleneck is carefully chosen, then the Moskowitz function of the approximation tends to the true function, $N^{(a)} \to N$, as $\Delta x \to 0$. [2] Thus, practical approximation schemes can be built.

Sections 2 and 3 below are preparatory. Section 2 describes how errors in the input data of KW moving bottleneck problems affect the Moskowitz solutions. Section 3 shows how errors due to imperfect bottleneck trajectories affect the solution—the errors can be reduced as much as desired

[1] A brief review of moving bottleneck theory can be found in Muñoz and Daganzo (2002).

[2] The Moskowitz function $N = N(t,x)$ (see Moskowitz, 1965, and Makigami et al., 1971) is the integral of flow that gives the vehicle number on the $(t,x)$ plane. Moskowitz functions are continuous with a bounded rate of variation, and are related to the density and flow functions by $\partial N/\partial x = -k$ and $\partial N/\partial t = q$.

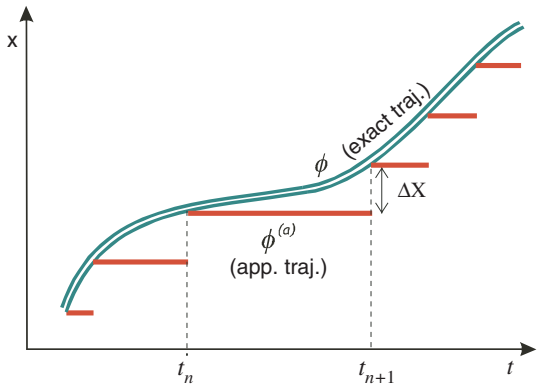



Fig. 1. Exact and approximate moving bottleneck trajectories.

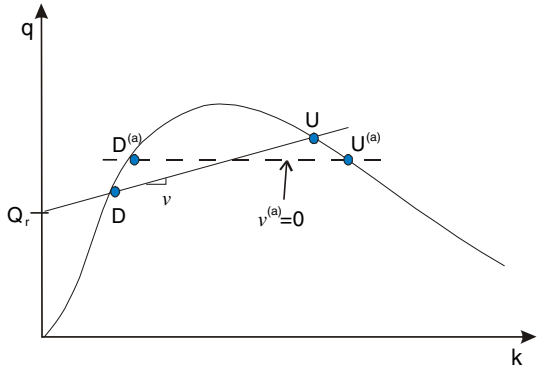



Fig. 2. Possible states next to an active moving bottleneck.

by reducing $\Delta x$. In Sections 2 and 3 time is continuous. Section 4 then shows how the results can be used to approximate the Moskowitz function with discrete-time numerical schemes. Section 5 discusses future work.

## 2. Error bounds: same boundary, different data

This section compares two KW solutions defined on the same boundary with different input densities along the boundary ("exact" and "approximate"). We start with conventional initial-value problems; then extend the results to more general boundary value problems, and finally to problems including moving bottlenecks. In all cases, it is shown that the solution error at any point $|N(a) - N|$ can never exceed the maximum input error.

### 2.1. Homogeneous initial-value problems

It is known (Lax, 1973) that the integral $\int_{-\infty}^{\infty} |k^{(a)}(t,x) - k(t,x)|\mathrm{d}x$ corresponding to any two density functions, $k^{(a)}(t,x)$ and $k(t,x)$, arising from the physically meaningful solution of two KW

problems with the same equilibrium flow-density relation, $F(k) \in C^1$, but different initial data is non-increasing in $t$. In other words, the separation of the two density profiles at a given time, as measured by the $L_1$ norm, cannot increase with $t$. This result is intimately related to the ideas of uniqueness and stability since it implies that solutions with the same input data must match everywhere (uniqueness), and also implies that small perturbations to initial data cannot grow into the solution (stability).

It turns out that the Moskowitz function satisfies a similar property with respect to the $L_\infty$ norm. It has been shown (see Eq. (6) in Daganzo, 2001, or (B6) in Daganzo, 2003) that if $F$ is continuous, non-negative and satisfies $q = 0$ for $k \notin [0, k_j]$, where $k_j$ is the "jam density", then $\sup_x |N^{(a)}(t,x) - N(t,x)|$ is non-increasing in $t$. [3] Thus, if we denote by "$E$" the absolute difference (i.e., the error) between the two Moskowitz functions at a given point, [4] we can write:

$$E \leqslant \sup_x [E(0,x)]. \tag{1a}$$

The same result can be expressed in a way that will be useful later if we call the curve defined by the Moskowitz function for a fixed value of $t$ an "$N$-profile" and denote it by $M_t$, and if we also define a kinematic wave operator $\mathscr{W}_t$ that returns $M_t$ from $M_0$; i.e., $M_t = \mathscr{W}_t M_0$. This operator is a property of $F$. If we now use double bars to denote the $L_\infty$ norm then the property in question becomes $\|\mathscr{W}_t M^{(a)} - \mathscr{W}_t M\| \leqslant \|M^{(a)} - M\|$, for any $M^{(a)}$, $M$ and $t$. In other words, the kinematic wave operator is a (non-strict) contraction mapping for any $t$. Operators with this property will be simply called "contractions". Note that for time-independent $F$-functions $\mathscr{W}_{t+t'} = \mathscr{W}_t \mathscr{W}_{t'}$. Thus, $\mathscr{W}_t$ can be expressed as the composition of many identical "differential" operators, $\mathscr{W}$. Since the composition of contractions is a contraction, an equivalent statement to (1a) is: "$\mathscr{W}$ is a contraction".

### 2.2. General boundaries

The contraction property also applies to more general boundary value problems. Let $C$ be a continuous curve on which two sets of well-posed density data have been defined, and let $P_0$ be a point $(t_0, x_0)$ in the range of influence of $C$; see Fig. 3. We now show that (1a) is generalized by:

$$E_0 \leqslant \sup_{R \in C} [E_R], \tag{1b}$$

where $E_0 = |N^{(a)}(t_0, x_0) - N(t_0, x_0)|$ and $E_R$ is the absolute difference in $N$-values for a point, $R$, on the curve.

The logic behind (1b) uses the fact that $E = |N^{(a)} - N|$ is a continuous function of $x$ and $t$ with a bounded rate of variation (since $N$ and $N^{(a)}$ are continuous with bounded rates of variation). Assume that (1b) is false; i.e., that there is an $\varepsilon > 0$ such that $E_0 \geqslant \varepsilon + E_R$, $\forall R \in C$. If we let $\Delta t$ be a small time increment, we know from (1a) that there must be a point $P_1$ in the domain of dependence of point $P_0$ on the line $t = t_0 - \Delta t$ such that $E_1 \geqslant E_0$. The same happens at $P_1$. Thus, we

[3] Actually, these references show that $\sup_t |N^{(a)}(t,x) - N(t,x)|$ is non-decreasing in $x$ for well-posed initial boundary problems where flows are given at $x = 0$. The argument of the references (with $t$ and $x$ reversed) applies to the initial value problem and shows that the result continues to hold in this case.

[4] The $t$ and $x$ arguments of functions such as $E$ will be omitted whenever possible in this paper.

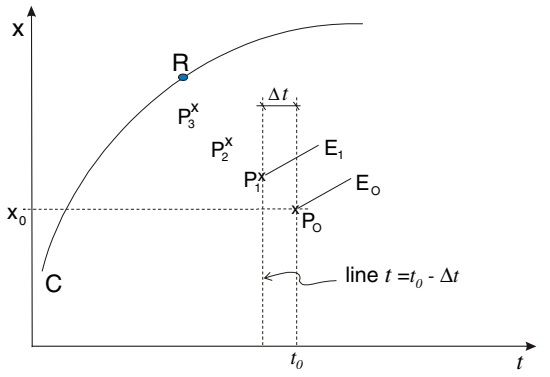



Fig. 3. Hypothetical problem with a curved boundary.

can form a sequence of points $\{P_i\}$, approaching the curve $C$, for which $E_i \geqslant E_0$; see Fig. 3. For all these points $E_i \geqslant E_0 \geqslant \varepsilon + E_R$; i.e., $E_i - E_R \geqslant \varepsilon$, $\forall R \in C$. This is independent of $\Delta t$. Since a point $P_i$ arbitrarily close to $C$ can always be found by reducing $\Delta t$, we see that $E$ cannot have a bounded rate of variation near $C$. Thus, (1b) must hold.

### 2.3. Problems with moving bottlenecks

First we need some definitions. From now on, flows and densities for the states (''$U$'' and ''$D$'') next to an active moving bottleneck with speed $v$ (see Fig. 2) will be denoted by capitalized letters and capitalized subscripts; e.g., $Q_D$, $K_U$. The downstream flow, $Q_D$, will be called ''the capacity'' of the moving bottleneck, and its subscript will be omitted. The capacity is exogenously defined. It can take any value and vary with time. Recall too, from the theory of moving bottlenecks, that the intercept of line $\overline{DU}$ with the $q$-axis of Fig. 2 is the maximum passing rate, $Q_r$.

We consider now moving bottlenecks with exogenous trajectories and capacities, and show that (1a) and (1b) continue to hold. A separate discussion is necessary because the bottleneck trajectory does not define a boundary condition of the type discussed in Section 2.2 since the densities along it are now endogenously determined—only its speed and capacity are exogenous. To establish the result we first modify the differential wave operator to include bottlenecks, and then show that it remains a contraction.

If a homogeneous freeway includes a moving bottleneck at $x = x_0$ and time $t$, then the $N$-profile upstream of the bottleneck at time $t + \mathrm{d}t$ can be obtained by applying the wave operator to the following two profiles at time $t$ (i.e., ignoring the bottleneck) and comparing the results: (i) the original $N$-profile, $M_t$, and (ii) an artificial profile, $M_t^{(U)}$, which is identical to $M_t$ except that it is in state ''$U$'' downstream of the bottleneck:

$$M_t^{(U)}(x) = M_t(x), \qquad \text{if } x \leqslant x_0, \tag{2a}$$

$$= M_t(x_0) - K_U(x - x_0) \quad \text{if } x > x_0. \tag{2b}$$

Application of the wave operator to (2) (case (ii)) gives the correct solution when the bottleneck is active. Application to $M_t$ (case (i)) gives the solution when it is not. The reader can verify by exhaustively examining all possibilities that the correct solution is always the lower of the two

cases. Thus, if we use an operator $\mathcal{U}$ to express (2), $M_t^{(U)} = \mathcal{U}M_t$, the solution upstream of the bottleneck is:

$$M_{t+\mathrm{d}t} = \min\{\mathcal{W}M_t; \mathcal{W}\mathcal{U}M_t\} \quad \text{for } x \leqslant x_0 + v\mathrm{d}t, \tag{3}$$

where the *min* operation denotes the lower envelope of the two curves. (For simplicity, our notation does not explicitly reflect the dependence of $\mathcal{U}$ on $x_0$.)

For the downstream part we define another location-dependent operator $\mathcal{D}$ that changes the upstream part of the profile:

$$M_t^{(D)}(x) = M_t(x), \qquad \text{if } x \geqslant x_0, \tag{4a}$$

$$= M_t(x_0) - K_D(x - x_0) \quad \text{if } x < x_0 \tag{4b}$$

and use a similar minimum rule:

$$M_{t+\mathrm{d}t} = \min\{\mathcal{W}M_t; \mathcal{W}\mathcal{D}M_t\} \quad \text{for } x \geqslant x_0 + v\mathrm{d}t. \tag{5}$$

The combination of (3) and (5) is the desired result. The complete operation will be denoted "$\mathcal{B}$", so that $M_{t+\mathrm{d}t} = \mathcal{B}M_t$.

Note as an aside that $M_{t+\mathrm{d}t}$ must be continuous since it is part of a KW solution. Reassuringly, the vehicle numbers predicted by both (3) and (5) at $x = x_0 + v\mathrm{d}t$ always match. As expected, these numbers never exceed $N_t(x_0) + Q_r\mathrm{d}t$, and they equal this value when the bottleneck is active. We now show that $\mathcal{B}$ is a contraction.

**Lemma 1.** *If* $(a, A, b, B)$ *are real numbers such that* $|A - B| \leqslant \varepsilon$ *and* $|a - b| \leqslant \varepsilon$, *then* $|\min(A, a) - \min(B, b)| \leqslant \varepsilon$.

**Proof.** If $(A, B) < (\text{or} >)(a, b)$ then $|\min(A, a) - \min(B, b)| = |A - B|$ or $|a - b|$, and the lemma is obviously true. Otherwise, when $(A, b) < (\text{or} >)(a, B)$, $|\min(A, a) - \min(B, b)| = |A - b|$ or $|B - a|$. By symmetry, it suffices to pursue the first of these two instances, checking the magnitude of $|A - b|$ when $A < a$ and $b < B$. If $A \geqslant b$, then $|A - b| \leqslant |a - b| \leqslant \varepsilon$. Otherwise, $|A - b| \leqslant |A - B| \leqslant \varepsilon$. Thus, $|A - b| \leqslant \varepsilon$ in both cases, and this concludes the proof. □

**Theorem 1.** *Operation* $\mathcal{B}$ *is a contraction.*

**Proof.** For the upstream part (3), we need to show that

$$\|M^{(a)} - M\| \leqslant \varepsilon \Rightarrow \|\min\{\mathcal{W}M^{(a)}; \mathcal{W}\mathcal{U}M^{(a)}\} - \min\{\mathcal{W}M; \mathcal{W}\mathcal{U}M\}\| \leqslant \varepsilon. \tag{6a}$$

Note first that (2) is a contraction. Therefore, $\mathcal{U}$ is a contraction. Since $\mathcal{W}$ is a contraction, $\mathcal{W}\mathcal{U}$ is a contraction too. Hence, if $\|M^{(a)} - M\| \leqslant \varepsilon$ then $\|\mathcal{W}M^{(a)} - \mathcal{W}M\| \leqslant \varepsilon$ and $\|\mathcal{W}\mathcal{U}M^{(a)} - \mathcal{W}\mathcal{U}M\| \leqslant \varepsilon$. Lemma 1 can now be invoked with: $A = \mathcal{W}M^{(a)}$, $B = \mathcal{W}M$, $a = \mathcal{W}\mathcal{U}M^{(a)}$, and $b = \mathcal{W}\mathcal{U}M$, and the result is (6a). Similar logic applies to the downstream part (5), and we find that

$$\|M^{(a)} - M\| \leqslant \varepsilon \Rightarrow \|\min\{\mathcal{W}M^{(a)}; \mathcal{W}\mathcal{D}M^{(a)}\} - \min\{\mathcal{W}M; \mathcal{W}\mathcal{D}M\}\| \leqslant \varepsilon. \tag{6b}$$

The combination of (6a) and (6b) establishes that $\mathcal{B}$ is a contraction, and this concludes the proof. □

Theorem 1 only establishes the desired result for the initial value problem, but the logic of Section 2.2 again reveals that (1b) holds. In fact, one can also see along similar lines of thought that the theorem and the inequality hold for general boundary problems with multiple bottlenecks.

## 3. Error bounds: different boundaries

We now use (1)–(6) to see how a moving bottleneck, $\phi(t)$ with capacity $Q(t)$, can be approximated by another one whose trajectory is a step function, $\phi^{(a)}$, defined on a spatial lattice $x_i = i\Delta x$. We assume that $\|\phi^{(a)} - \phi\| \leqslant \Delta x$ and $\phi^{(a)} \leqslant \phi$ as in Fig. 1.

It will be shown in this section that if $Q^{(a)}(t) = Q(t)$ for all $t$ then $N^{(a)} \to N$ as $\Delta x \to 0$. We will prove this assertion first for piecewise linear problems, where $F(k)$, $N(0,x)$ and $\phi$ are piecewise linear, and $Q(t)$ is piecewise constant. Piecewise linear problems are easy to treat because their solutions, $k(t,x)$, are piecewise constant in a polygonal tessellation of the $(t,x)$ plane. The edges of these polygons are "interfaces" that satisfy the "shock condition". The vertices of the polygon can be associated with a sequence of times, $\{\tau_p\}$; see Fig. 4.

If the initial data on the line $t = 0$ includes a finite number of constant-density sates, and the moving bottleneck is also composed of a finite number of linear segments with constant $Q$, then the tessellation has a finite number of vertices and the series $\{\tau_p\}$ where interfaces converge or diverge has a finite number of elements. [5]

Lemmas 2 and 3, below, show that if $|N^{(a)} - N|$ is bounded by $\varepsilon_p$ at time $t = \tau_p$ then $|N^{(a)} - N|$ is bounded by $\varepsilon_p + 3k_j\Delta x$ in the interval $(\tau_p, \tau_{p+1})$. Thus, if we choose $\varepsilon_0 = 0$, we see by iteration of this result for $p = 0, 1, \ldots, P$ that $|N^{(a)} - N| \leqslant 3pk_j\Delta x$ in every interval $[\tau_p, \tau_{p+1}]$, and since the number of intervals is finite, this establishes that $N^{(a)} \to N$ uniformly in $(t,x)$; i.e., that the following is true:

**Theorem 2.** *For piecewise linear problems with P vertices, if we choose* $Q^{(a)}(t) = Q(t)$ *and* $\phi^{(a)}$ *is of the form shown in Fig.* 1, *then* $|N^{(a)} - N| \leqslant 3Pk_j\Delta x$. □

We now turn our attention to the Lemmas. Let $\{t_n\}$ be the set of instants where $\phi^{(a)}$ has a step. We examine first the propagation of errors in stationary intervals that begin with a step, and then relax the "step condition".

**Lemma 2.** *If* $\tau_p \in \{t_n\}$ *and* $|N^{(a)} - N| \leqslant \varepsilon_p$ *for all x and* $t = \tau_p$, *then* $|N^{(a)} - N| \leqslant \varepsilon_p + k_j\Delta x$ *for all x and* $t \in [\tau_p, \tau_{p+1}]$.

[5] This should be intuitive if $F$ is concave and the bottleneck eventually leaves the road because, in the concave case, non-bottleneck vertices eliminate interfaces. Thus, once the bottleneck has left the road the number of interfaces declines at each $\tau_p$. The process stops (after a finite number of steps) when one interface remains, or before if all the interfaces diverge. More elaborate geometrical considerations show that finite stopping also occurs in the non-concave case, even if the bottleneck stays on the road.

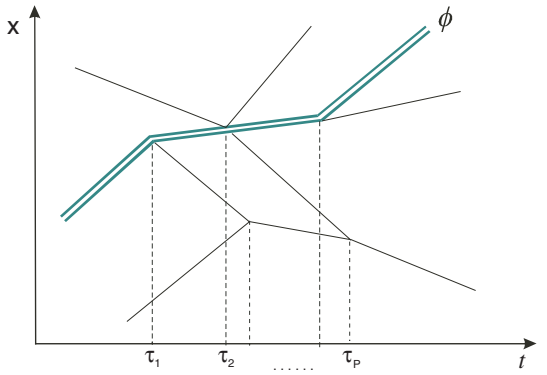



Fig. 4. Typical solution of a piece-wise linear problem.

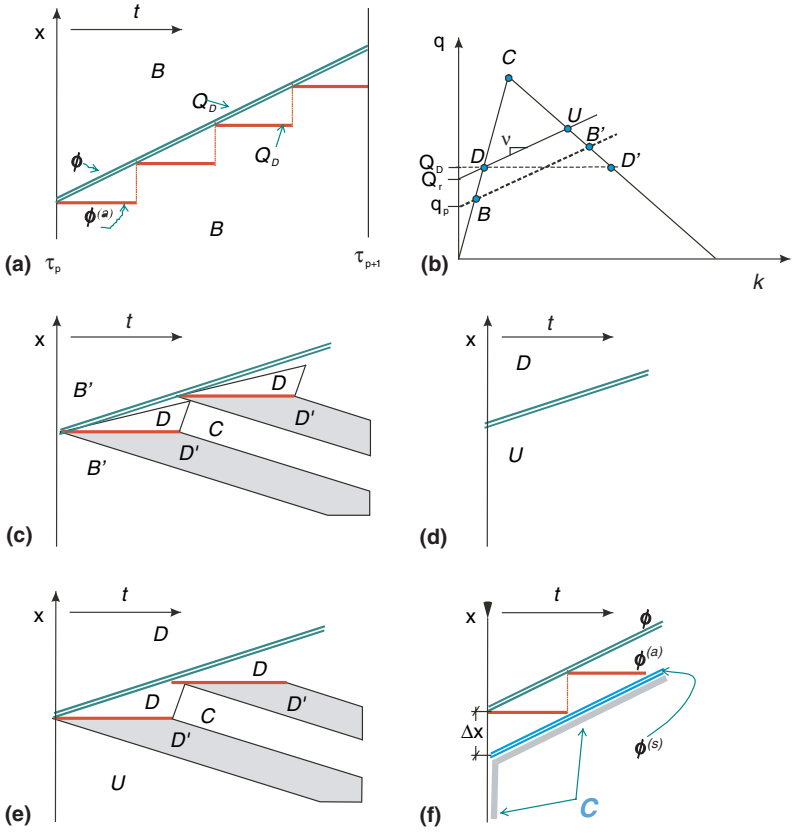



Fig. 5. Relevant diagrams when $\phi$ and $\phi^{(a)}$ intersect at $t = \tau_p$.

**Proof.** Consider the two bottleneck trajectories of Fig. 5(a) and note that they intersect at $t = \tau_p$, as required by the lemma. Let $N^*$ be the solution for a problem with the same initial data as the exact problem (i.e., such that $N^* = N$ for $t = \tau_p$) but the approximate bottleneck. We call this solution the "test" solution and shall compare it with $N$ and $N^{(a)}$.

For $t \in [\tau_p, \tau_{p+1}]$ the bottleneck must be in a steady state with a constant passing rate $q_p$. This rate must satisfy $q_p \leqslant Q_r$, where $Q_r$ is the maximum possible passing rate. Equality means that the bottleneck is active, but the bottleneck may or may not be active. If the bottleneck is inactive the same traffic state should be observed upstream and downstream of it. Fig. 5(b) shows the two states ($B$ or $B'$) associated with a specific $q_p < Q_r$. Recall, however, that if the bottleneck is active the steady states will be different on its upstream and downstream sides ($U$ and $D$).

Consideration of the test solution for all possible steady states reveals that $N^*$ always equals $N$ downstream of $\phi$. Differences between the two solutions only arise in the region upstream of $\phi$. For example, if the bottleneck were inactive in state $B'$ as in Fig. 5(b) the exact solution would consist of state $B'$ on both sides of the bottleneck. The test solution, however, would be as in Fig. 5(c), which induces states other than $B'$—but only upstream of $\phi$.

If the bottleneck was active then the exact solution would be as in Fig. 5(d), but the test solution would be as in Fig. 5(e); again, with no changes downstream of $\phi$. This is always true. Thus, $N = N^*$ for $x \geqslant \phi$. An observer moving in front of the bottleneck would never know whether the true or approximate bottleneck was in force.

Consider now the region between $\phi$ and the shifted curve $\phi^{(s)} = \phi - \Delta x$, which includes $\phi^{(a)}$; see Fig. 5(f). Since the width of this region is $\Delta x$, and $N = N^*$ along its top edge, it follows that $|N^* - N| \leqslant k_j \Delta x$ in the region.

This bound also holds beneath $\phi^{(s)}$, because in this region the solution is determined by data on the boundary $C$ (see Fig. 5(f)), and in these cases (see (1b)) the discrepancy in the two solutions is bounded by the maximum discrepancy along $C$, which is itself bounded by $k_j \Delta x$. Thus $|N^* - N| \leqslant k_j \Delta x$ for $t \in [\tau_p, \tau_{p+1}]$.

If Theorem 1 is now applied to $N^{(a)}$ and $N^*$ with the line $t = \tau_p$ as the boundary, where the error is bounded by $|N^{(a)} - N^*| \equiv |N^{(a)} - N| \leqslant \varepsilon_p$, we find that $|N^{(a)} - N^*| \leqslant \varepsilon_p$ for $t \in [\tau_p, \tau_{p+1}]$. Thus, $|N^{(a)} - N^*| + |N^* - N| \leqslant \varepsilon_p + k_j \Delta x$ for $t \in [\tau_p, \tau_{p+1}]$, and the triangle inequality $|N^{(a)} - N| \leqslant |N^{(a)} - N^*| + |N^* - N|$ ensures that $|N^{(a)} - N| \leqslant \varepsilon_p + k_j \Delta x$, as claimed. □

This lemma assumed that $\tau_p \in \{t_n\}$ but the result can be generalized.

**Lemma 3.** *If* $|N^{(a)} - N| \leqslant \varepsilon_p$ *for all* $x$ *and* $t = \tau_p$, *then* $|N^{(a)} - N| \leqslant \varepsilon_p + 3k_j \Delta x$ *for all* $x$ *and* $t \in [\tau_p, \tau_{p+1}]$.

**Proof.** In this case there is a discrepancy, $\delta = \phi - \phi^{(a)} > 0$ at $t = \tau_p$; see Fig. 6. We consider the auxiliary shifted solution, $N^{(s)}(x,t) = N(x + \delta, t)$, formed by shifting the exact bottleneck and the exact data by $\delta$ space units. Note that the errors induced by the shift are bounded by $|N^{(s)} - N| \leqslant k_j \delta$. We consider this solution in the interval $[\tau_p, \tau_{(p+1)}]$, where $\tau_{(p+1)}$ is the first time the shifted trajectory intersects either a step of $\phi^{(a)}$ (as shown in the figure) or the line $t = \tau_{p+1}$. [6]

We first estimate the error $|N^{(a)} - N^{(s)}|$ in that interval by introducing an auxiliary approximate bottleneck $\phi^{(s)}$ with steps at $\tau_p$ and $\tau_{(p+1)}$ that matches $\phi^{(a)}$ in $[\tau_p, \tau_{(p+1)}]$; see the dashed horizontal line in the figure. This bottleneck satisfies the step condition of Lemma 2. Thus, the lemma can be

[6] In other words, $\tau_{(p+1)}$ is the first member of $\{t_n\}$ in the interval $(\tau_p, \tau_{p+1}]$, if such a member exists, or else it is $\tau_{p+1}$.

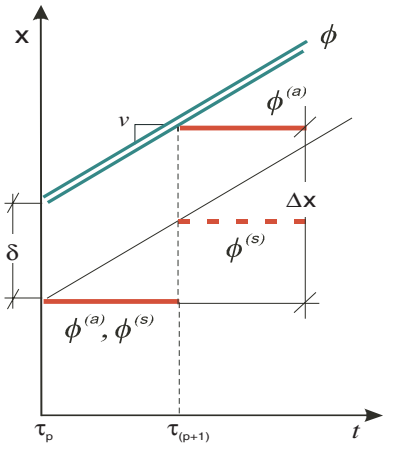



Fig. 6. Shifted solution when $\phi$ an $\phi^{(a)}$ do not intersect at $t = \tau_p$.

used to compare $N^{(a)}$ and $N^{(s)}$ in $[\tau_p, \tau_{(p+1)}]$. Since the step size of $\phi^{(s)}$ is bounded by $(\Delta x - \delta)$, see figure, Lemma 2 implies:

$$|N^{(a)} - N^{(s)}| \leqslant \varepsilon_{a,s} + k_j(\Delta x - \delta) \quad \text{for } t \in [\tau_p, \tau_{(p+1)}], \tag{7}$$

where $\varepsilon_{a,s}$ is the maximum error between $N^{(a)}$ and $N^{(s)}$ at time $\tau_p$. Since $|N^{(a)} - N^{(s)}| \leqslant |N^{(a)} - N| + |N - N^{(s)}|$ everywhere, the supremum of the left side of this inequality along the initial boundary, $\varepsilon_{a,s}$, is bounded from above by the supremum of the right side. And since the right side is itself bounded by $\varepsilon_p + k_j\delta$, we see that $\varepsilon_{a,s} \leqslant \varepsilon_p + k_j\delta$. If this inequality is now introduced in the right side of (7), we find:

$$|N^{(a)} - N^{(s)}| \leqslant \varepsilon_p + k_j\Delta x \quad \text{for } t \in [\tau_p, \tau_{(p+1)}]. \tag{8}$$

We can now estimate the total error. Since $|N^{(s)} - N| \leqslant k_j\delta$ for $t \in [\tau_p, \tau_{(p+1)}]$ we can use the triangle inequality and (8) to write:

$$|N^{(a)} - N| \leqslant |N^{(a)} - N^{(s)}| + |N^{(s)} - N| \leqslant \varepsilon_p + k_j\Delta x + k_j\delta \leqslant \varepsilon_p + 2k_j\Delta x \quad \text{for } t \in [\tau_p, \tau_{(p+1)}].$$

Obviously, if $\tau_{(p+1)} = \tau_{p+1}$, the proof is completed. Otherwise, $\phi^{(a)}$ has a step at time $\tau_{(p+1)}$, as in Fig. 6, and Lemma 2 can be used to estimate the error in $[\tau_{(p+1)}, \tau_{p+1}]$. Since the revised input error is $\varepsilon_p + 2k_j\Delta x$ at time $\tau_{(p+1)}$, the resulting bound is $\varepsilon_p + 3k_j\Delta x$. □

This lemma is the last step in the proof of Theorem 2, which pertains to single-bottleneck, piece-wise linear problems. These restrictions can be relaxed as follows.

First note that if one allows $\phi$ and $\phi^{(a)}$ to represent sets of trajectories for multiple bottlenecks, rather than individual trajectories, and we also let the capacity of $\phi^{(a)}$ when bottlenecks share the same location be the minimum of their $Q$'s, then it becomes apparent from the foregoing logic that the result also holds for problems with multiple bottlenecks. [7]

[7] Other capacity models can also be used for overlapping bottlenecks, but the particular model is irrelevant if bottlenecks spend very little time next to each other.

Bounds can also be obtained for the class of problems whose input data can be approximated arbitrarily well by a sequence of piecewise linear problems. In this case, if we let $P_i$ denote the number of vertices for the $i$th problem and approximate its solution using a bottleneck step $(\Delta x)_i = (\Delta x)_0/(iP_i)$ we see from the theorem that the error bound for the $i$th problem is $3(\Delta x)_0 k_j/i$, which tends to zero as $i \to \infty$.

The proposed approach also works for continuous $F$-functions. To see this, we recall from Daganzo (2001) that the error in the solution introduced by using a piecewise linear approximation $F^{(a)}$ to $F$, instead of $F$, is of order $\Delta q$, where $\Delta q = \sup|F^{(a)} - F|$. The rationale for our claim is the following. Since $\Delta q$ can be made as small as desired by increasing the number of pieces in $F^{(a)}$, we see that the piecewise solution tends to the continuous solution as $\Delta q \to 0$, both with the exact and the step bottlenecks. Hence, since the step solution can be as close as desired to the exact in the piecewise linear case for any $\Delta q$, it follows that the same must happen in the continuous case.

The same argument reveals that the method can also be used with inhomogeneous highways (and networks) since they can always be approximated arbitrarily well by combination of homogeneous pieces.

## 4. Discrete-time approximation: examples

The above results guarantee that *any* finite difference (discrete-time) approximation that converges to the true solution of kinematic wave problems with fixed bottlenecks such as Newell's exact method for triangular $F$-functions (Newell, 1993) can also be successfully used without modification with moving bottlenecks. All one has to do is choose a sufficiently small $\Delta x$ to approximate the bottleneck trajectory, and an even smaller lattice spacing for the finite difference scheme. In practice, however, one can choose both lattice spacings to be equal with acceptable results. Two examples are now given.

**Example 1** (*Accuracy of the procedure*). As an illustration we now solve with an "off-the shelf" version of the cell transmission model a problem where a vehicle embedded in traffic suddenly slows down, and then compare the results with the exact solution of KW theory. We examine a one-mile homogeneous section of a freeway, whose $F$-function is an isosceles triangle with free flow speed, $v_f = 60$ mph, jam density, $k_j = 300$ veh/mi, and wave velocity in congestion, $w = -v_f$. Although the isosceles shape is not realistic (it has a maximum flow of 9000 vph), it is useful because the CT model does not introduce any numerical error for isosceles triangles if we choose $\Delta x/\Delta t = v_f$ (Daganzo, 1994). Thus, the discrepancies between exact and approximate solutions about to be presented can be exclusively attributed to the approximation method of this paper. We assume that the freeway is flowing at capacity when, at $t_0 = 0.3$ min and $x = 0.3$ mi, a vehicle instantaneously slows down to a constant speed $v = 20$ mph and then resumes free-flow speed at $t = 2.1$ min. We assume that the moving restriction halves the capacity; i.e., $Q = 4500$ vph. Fig. 7 shows the solution of this problem on the $(t,x)$ plane. Capital letters indicate the states observed in the shown regions of the $(t,x)$-plane—"$C$" stands for freeway capacity and, as before, "$D$ and $U$", for the states next to the active bottleneck. This part of the figure also shows a set of detectors

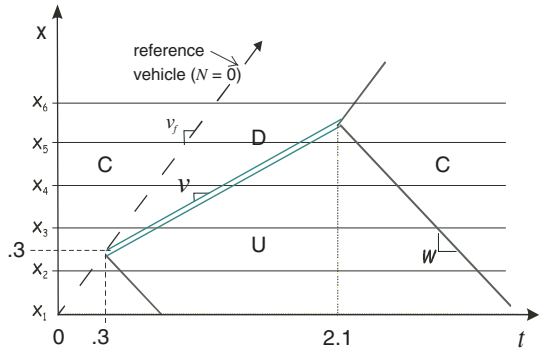



Fig. 7. Exact $(t,x)$ solution of a moving bottleneck example.

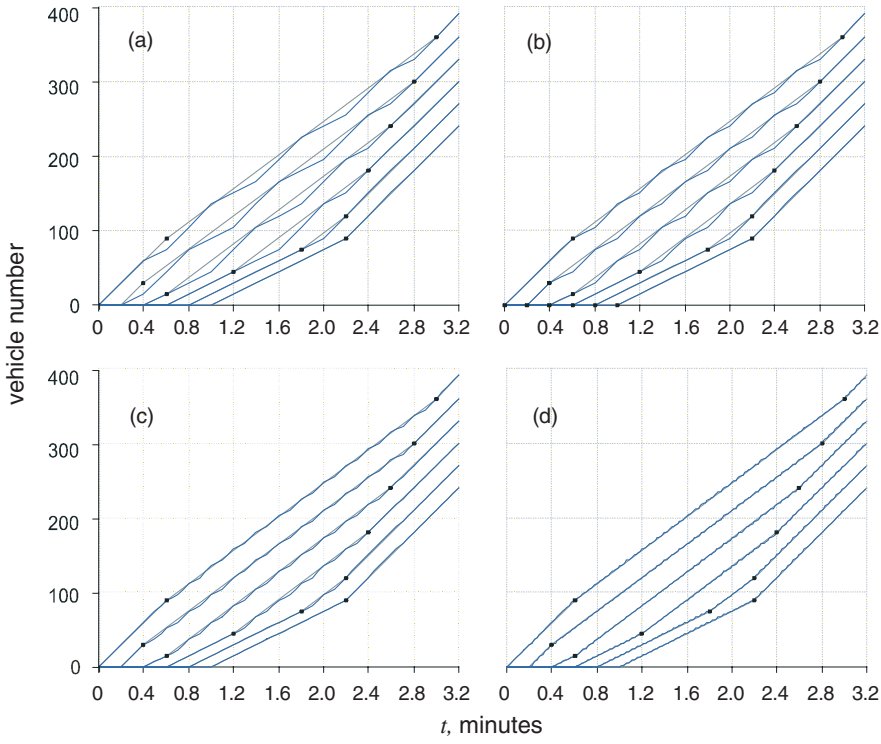



Fig. 8. Numerical solutions for the example for different mesh sizes (in seconds): (a) $\Delta t = 12$; (b) $\Delta t = 6$; (c) $\Delta t = 3$; (d) $\Delta t = 1$.

at locations $\{x_i\}$, spaced 1/5 mi apart. The exact $N$-values observed at these detectors, $N(t,x_i)$, will be used for the comparisons, using $N = 0$ for the last vehicle unaffected by the obstruction; see figure.

Four numerical solutions were obtained with the following meshes ($\Delta t$ s, $\Delta x$ mi) = (12, 1/5); (6, 1/10); (3, 1/20) and (1, 1/60). In all cases $\Delta x/\Delta t = v_f$, to ensure that the CT results do not contain numerical errors. Fig. 8 shows the four numerical solutions, overlaid over the exact solution. The

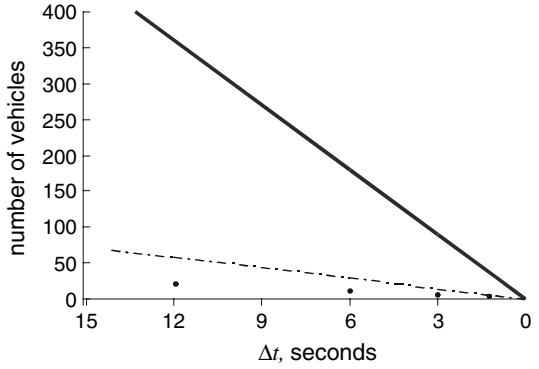



Fig. 9. Actual errors for the example (dots) with the upper bounds arising from Theorem 2 (solid line) and Lemma 2 (dashed line).

black dots are the interfaces between traffic states in the exact solution. Note how the discrepancies disappear, as the mesh is refined and the very high resolution that can be achieved. Fig. 9 shows the maximum discrepancy for each case by means of a dot, and uses a solid line for the bound of Theorem 2, for $P = 2$, which is amply met. The exact discrepancies are 22.50, 11.25, 5.63 and 1.88 vehicles for $\Delta t = 12$, 6, 3 and 1 s, respectively.

For problems with only one queuing episode and no error in the initial data, such as ours, one can use directly the (tighter) bound of Lemma 2, $k_j \Delta x$. This is also shown in the figure by means of a dashed line. This could be useful if one wished to estimate the exact error for a particular problem.

Note in closing that practical measures of performance for traffic streams are always defined over finite regions of the $(t,x)$-plane and that they can usually be evaluated if one knows the Moskowitz function in the region of interest. This is certainly the case for average flows, densities, vehicle-miles, vehicle-hours, and any other measures of vehicular accumulation, delay or travel time, as shown by the formulas in Section 4.1.1 of Daganzo (1997). Furthermore, practical measures of performance pertaining to finite regions of space-time always turn out to be continuous functionals of the Moskowitz function; see for example Eq. (4.6) in Daganzo (1997). Therefore, an error as small as desired for a measure of performance can always be obtained by reducing the error in the Moskowitz function; i.e., by reducing $\Delta x$ and using the procedure proposed in this paper.

**Example 2** (*Two bottlenecks*). This example shows how the method handles bottleneck passing. Subscripts "1" and "2" are used to label bottleneck-specific parameters. We continue to consider the freeway of Example 1, described by the flow-density diagram of Fig. 10(a), but now assume that the freeway is empty at $t = 0$, except for a lone truck at $x = 0.3$ mi; see Fig. 10(b). The truck travels at a constant speed, $v_1 = 11$ mph, and reduces the capacity by a factor of two; i.e., $Q_1 = 4500$ vph. At this time, a police car traveling at $v_2 = 45$ mph enters the road at $x = 0$, followed by a queue of cars. We assume that no cars pass the police car; i.e., $Q_2 = 0$. Since the flow within the queue of cars trailing the police car (state $U_2$) is higher than $Q_1$, queued vehicles will not be able to pass the truck unhindered after the police car overtakes it—a residual queue must form behind the

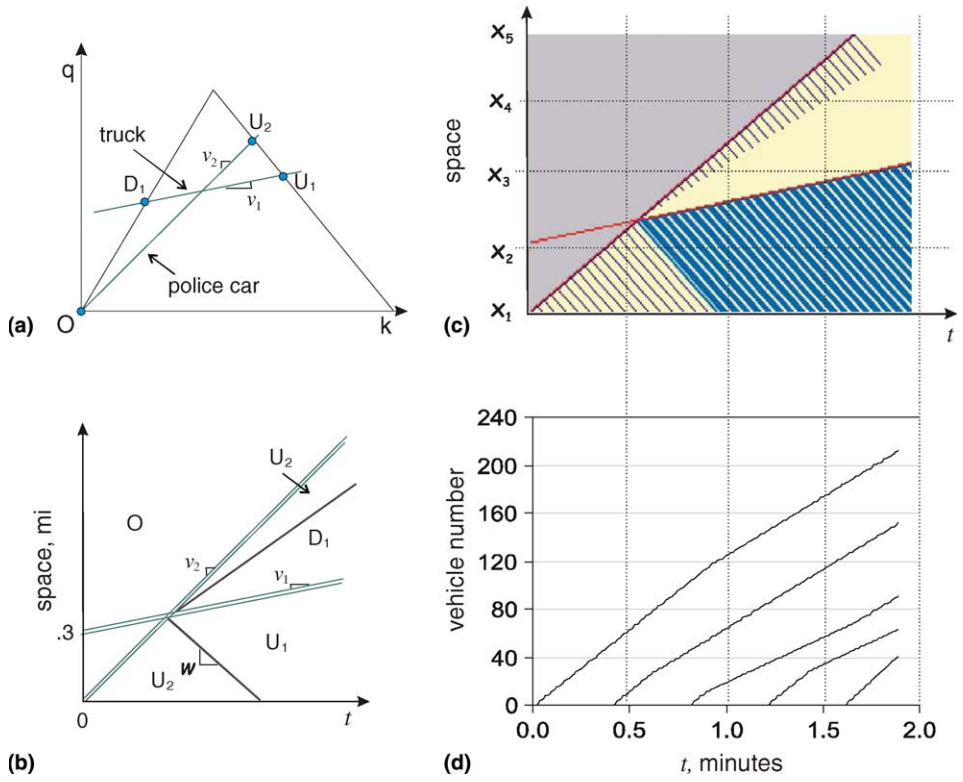



Fig. 10. Solution of Example 2: (a) and (b) exact solution; (c) and (d) numerical solution with $\Delta t = 1$ s.

truck. The precise details of this process are displayed in Fig. 10, which contains the exact solution to the problem. The same queuing pattern arises in the numerical solution, shown in Fig. 10(c). Note the perfect match between the two diagrams. Fig. 10(d) displays the $N$-curves obtained with the numerical method. The exact $N$-curves are not included because the discrepancies between the two sets of curves are too small for the eye to see.

The method also works with variable bottleneck speeds, and this is illustrated in Fig. 11. The only difference between the problems of Figs. 10 and 11 is that in the latter case the two bottlenecks start at $t = 0$ with speeds $v_1 = 0$ mph and $v_2 = 40$ mph, and they accelerate.

## 5. Discussion and future work

This paper has examined the effects of moving bottlenecks on traffic streams, but not the reverse. A model with the reverse effects is necessary if one wishes to treat bottlenecks as endogenous components of a traffic stream; e.g., to characterize its behavior when it contains under-performing vehicles such as truck and buses.

The proposed method overcomes the main limitations of existing methods, and has been adapted to the endogenous case by introducing a car-following model for the slow vehicles (Laval

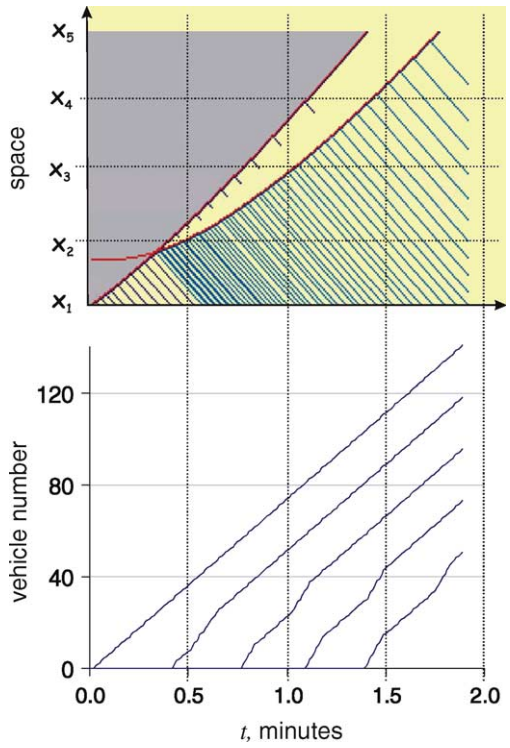



Fig. 11. Numerical solution of Example 2 with acceleration; $\Delta t = 1$ s.

and Daganzo, 2004). The car-following model uses as a stimulus (input) the speed of the traffic stream averaged over a fixed distance ahead of the bottleneck, and also includes as an input the status of other nearby bottlenecks. The averaging is necessary to remove the fictitious fluctuations in speed generated by the proposed method inside queues.

An extension of Godunov's method that produces solutions with smooth speeds in cases more general than those analyzed in Giorgi et al. (2002) has been developed in Daganzo and Laval (2003). This method does not require averaging but exhibits Godunov's (first order) numerical errors and cannot be used with "off the shelf" software.

The results in this paper also allow us to model lane-changes on multi-lane freeways as randomly occurring moving bottlenecks in a single-pipe KW stream. The output of such a model should qualitatively match the observations in Mauch and Cassidy (2002), since lane-changes in such a model tend to increase oscillation amplitudes without changing how they propagate—as observed. This hybrid theory can also help explain the puzzling relationship between merge capacity and on-ramp flow measured in Cassidy and Rudjanakanoknad (2002). This is explained in Laval (2004).

It is also possible to extend the hybrid modeling ideas in this paper to multi-pipe models of traffic flow. This has the appeal that an exogenous random capacity does not have to be defined for each lane change, although one needs to formulate instead a model of inter-lane interaction. This possibility is explored in Laval (2004).

## References

Cassidy, M.J., Rudjanakanoknad, J., 2002. Empirical Study of Ramp Metering and Capacity. Research Report UCB-ITS-RR-2002-5, University of California, Berkeley.

Daganzo, C.F., 1994. The cell transmission model: A dynamic representation of highway traffic consistent with the hydrodynamic theory. Trans. Res. B 28, 269–287.

Daganzo, C.F., 1997. Fundamental of Transportation and Traffic Operations. Elsevier, New York, NY.

Daganzo, C.F., 2001. A simple traffic analysis procedure. Nets. Sptl. Econ. 1, 77–101.

Daganzo, C.F., 2003. A Theory of Supply Chains. Springer, Heidelberg, Germany.

Daganzo, C.F., Laval, J., 2003. Moving bottlenecks: A numerical method that converges in flows. Institute of Transportation Studies Research Report UCB-ITS-RR-2003-02, University of California, Berkeley, California.

Gazis, D.C., Herman, R., 1992. The moving and 'phantom' bottlenecks. Trans. Sci. 26, 223–229.

Giorgi, F., Leclercq, L., Lesort, J.B., 2002. A traffic flow model for urban traffic analysis: Extensions of the LWR model for urban and environmental applications. In: Taylor, M.A.P. (Ed.), Proc. 15th ISTTT. Pergamon, Adelaide, Australia, pp. 393–415.

Koshi, M., Kuwahara, M., Akahane, H., 1992. Capacity of sags and tunnels in Japanese motorways. ITE J. (5), 17–22.

Laval, J.A., 2004. Hybrid Models of Traffic Flow: Impacts of Bounded Vehicle Accelerations, Ph.D. thesis. Department of Civil and Environmental Engineering, University of California, Berkeley.

Laval, J.A., Daganzo, C.F., 2004. A Hybrid Model of Traffic Flow: Impacts of Roadway Geometry on Capacity. Presented at the 2004 meeting of the Trans. Res. Board (submitted for publication).

Lax, P.D., 1973. Hyperbolic systems of conservation laws and the mathematical theory of shockwaves, SIAM Regional Conference Series in Applied Mathematics, J.W. Arrowsmith Ltd., Bristol, UK.

Lebacque, J.P., Lesort, J.B., Giorgi, F., 1998. Introducing buses into first order macroscopic traffic flow models. Trans. Res. Rec. 1644, 70–79.

Lighthill, M.J., Whitham, G.B., 1955. On kinematic waves. I Flow movement in long rivers. II A theory of traffic flow on long crowded roads. Proc. Roy. Soc. A 229, 281–345.

Makigami, Y., Newell, G.F., Rothery, R., 1971. Three-dimensional representation of traffic flow. Trans. Sci. 5, 302–313.

Mauch, M., Cassidy, M.J., 2002. Freeway traffic oscillations: Observations and predictions. In: Taylor, M.A.P. (Ed.), Proc. 15th ISTTT. Pergamon, Adelaide, Australia, pp. 653–674.

Moskowitz, K., 1965. Discussion of freeway level of service as influenced by volume and capacity characteristics. In: Drew, D.R., Keese, C.J. (Eds.), Highway Res. Rec. 99, 43–44.

Muñoz, J.C., Daganzo, C.F., 2002. Moving bottlenecks: A theory grounded on experimental observation. In: Taylor, M.A.P. (Ed.), Proc. 15th ISTTT. Pergamon, Adelaide, Australia, pp. 441–462.

Newell, G.F., 1993. A simplified theory of kinematic waves in highway traffic, I general theory, II queuing at freeway bottlenecks, III multi-destination flows. Trans. Res. B 27, 281–313.

Newell, G.F., 1998. A moving bottleneck. Trans. Res. B 32, 531–537.

Richards, P.I., 1956. Shockwaves on the highway. Oper. Res. 4, 42–51.